Gen-3 Theoretical Framework: Full Neural Replacement of the LHCb Runge–Kutta Track Extrapolator

TrackExtrapolation / gen-3
consolidated 2026-05-19

2026-05-19

Abstract

The canonical project goal, stated verbatim in PROJECT_CONTEXT.md, is to *"replace the C++ adaptive Runge–Kutta track extrapolator with a faster neural network that maintains high accuracy."* As of 2026-05-19 the gen-3 programme is re-anchored on this *replacement* criterion: at inference time the candidate must be the entire map from initial to final state, with no analytic Lorentz right-hand side and no field-map lookup. Under this criterion all hybrid candidates — legacy `PINN`, `RK_PINN` and `NeuralRK4` — are demoted, and only two architecture families survive: the plain `MLP` (architectures.py line 178) and `PINN_v2` (*ibid.* line 232). Three blockers stand between these survivors and Stage-1 deployment: a metric pathology (**Fix L1**, heavy-tail-dominated MSE), an MLP input-layout regression (**Fix M1**, missing z_0 and signed-Δz features) and an unscaled PINN_v2 (**Fix P1**, parameter-starved at $\sim$10 k weights). The six-phase R1–R6 roadmap of REPLACEMENT_PLAN.md addresses each blocker in dependency order, with an R-X contingency for the open A4 (Jacobian) question. The empirical companion document gen3_replacement_results_2026-05-19.tex reports the corresponding measurements.

1 The replacement criterion

Definition. A trained model f_θ is a *replacement* of the C++ adaptive Runge–Kutta extrapolator iff, at *inference* time, the only inputs are the 7-tuple

$$\mathbf{x} \;=\; (x,\ y,\ t_x,\ t_y,\ q/p,\ z_0,\ \Delta z) \in \mathbb{R}^7, \tag{1}$$

the only outputs are the 5-tuple

$$f_\theta(\mathbf{x}) \;=\; (x_f,\ y_f,\ t_{x,f},\ t_{y,f},\ q/p_f) \in \mathbb{R}^5, \tag{2}$$

and the computational graph of f_θ contains *no* evaluation of the analytic Lorentz right-hand side

$$f_{\mathrm{Lorentz}}(\mathbf{y}, z;\, q/p) \;=\; \begin{pmatrix} t_x \\ t_y \\ \kappa\, N \left[\, t_y(t_x B_x + B_z) - (1+t_x^2) B_y \,\right] \\ \kappa\, N \left[\, -t_x(t_y B_y + B_z) + (1+t_y^2) B_x \,\right] \end{pmatrix}, \qquad N = \sqrt{1 + t_x^2 + t_y^2}, \tag{3}$$

and *no* lookup into the LHCb dipole field map $\mathbf{B}(x, y, z)$ (stored as a $81 \times 81 \times 146$ trilinear grid in `twodip.rtf`; PROJECT_CONTEXT.md §"Physics").

Hybrids violate the criterion. A model that runs a classical RK4 loop over f_{Lorentz} with a small learned correction is an *augmentation* of the analytic integrator, not a replacement: every sub-step calls $\mathbf{B}(\cdot)$ and evaluates (**??**). The inference graph carries the entire field map as a dependency and inherits the classical integrator's cost structure. The hybrid may be accurate, but it cannot remove the production bottleneck the project was set up to remove (PROJECT_CONTEXT.md §"The Problem"); it can at best speed it up by a constant factor through step-count reduction.

What this rules in and out. REPLACEMENT_PLAN.md §1 and ADR 0009 (0009-replacement-goal-restated.m formalise three things that are *not* replacements: (i) a small neural correction added to an RK4 base prediction; (ii) a learned right-hand side evaluated inside a classical RK4 loop (Neural-ODE flavour); (iii) a classical RK4 with a single trainable hyperparameter. Each evaluates (**??**) at inference time. Each is therefore retired from the deployment candidate list.

2 Architecture taxonomy

Table **??** classifies the five families that have been trained in the project according to the replacement criterion of Section **??**. The verdicts are taken from REPLACEMENT_PLAN.md §2 and the gen-2 audit ISSUES_WITH_PINNS_AND_RK_PINNS.md §§3, 5.

Table 1: Architecture families and their replacement-criterion verdict. "Inference cost" lists the dominant per-call operation. "Replacement?" applies the Section **??** definition.

Family	Inference cost	Replacement?	Justification
`MLP`	single matmul chain	✓	Direct map $\mathbb{R}^7 \to \mathbb{R}^5$; no field, no f_{Lorentz}. Class at `architectures.py` line 178.
`PINN_v2`	single matmul chain (at $\zeta = 1$)	✓	Encoder of $[\mathbf{x}_0, \zeta]$ with the field consumed only in the *training* PDE residual; at inference $\zeta = 1$ and the network alone is evaluated. Class at `architectures.py` line 232.
legacy `PINN`	matmul + structural floor	×	Trunk is z-independent $\Rightarrow$ trajectory affine in ζ; PDE residual has a structural floor and IC loss is identically zero by construction (ISSUES §3). Superseded by `PINN_v2`.
legacy `RK_PINN`	4 head matmuls + softmax	×	Softmax mixture of *states* at four ζ values; not an RK4 quadrature of *derivatives*. Backbone detached from physics by `features.detach()`. Slope error 43–587× worse than MLP (ISSUES §5).
`NeuralRK4`	N_{RK} Lorentz RHS + matmul correction	×	Calls f_{Lorentz} and $\mathbf{B}(\cdot)$ at inference. A hybrid, not a replacement (`CLEANUP_LIST.md` §3.1). Demoted to research baseline.

Reasoning chain for the hybrid demotion. In gen-2 the best hybrid (`neural_rk4_small_1step`, 17.9 k params, 0.125 mm) outperformed the best true replacement (`pinn_v2_medium_lam0.1`, 68.6 k params, 0.235 mm) by roughly 2× (PROJECT_CONTEXT.md §"Current State"). On the gen-3 signed-Δz data contract the gap closed: the best true replacement `pinn_v2_small_v1` (10.4 k params, 0.117 mm) *matched* the best hybrid `nrk4_tiny_1step_v1` (5.0 k params, 0.113 mm) to within statistical noise (`REPLACEMENT_PLAN.md` §3). Combined with the canonical goal mandating replacement, the hybrid is therefore retired: there is no longer a measured accuracy advantage that would justify accepting a candidate that fails the criterion of Section **??**.

3 Loss-metric reform (Fix L1)

Pathology. The gen-3 M1 audit (`gen_3/README.md` §F2) reports for a single trained model with a single data pipeline

$$\mathcal{L}^{\text{train}} = 2.162 \times 10^{-4}, \quad \mathcal{L}^{\text{val}} = 1.426 \times 10^{-5}, \quad \mathcal{L}^{\text{test}} = 1.918 \times 10^{-7}, \tag{4}$$

a spread of $\sim 1100\times$ across splits drawn from the same distribution. The split-to-split variance is entirely driven by which split happens to contain the worst $<1\%$ of tracks.

Why squared residuals are unstable here. Write the per-track residual on output channel k as $r_k = (y_{\theta,k} - y_k^\star)/\sigma_k$, where σ_k is the output standard deviation used for non-dimensionalisation in ISSUES §1. The mini-batch MSE is

$$\mathcal{L}_{\text{MSE}} \;=\; \frac{1}{NK} \sum_{n=1}^{N} \sum_{k=1}^{K} r_{n,k}^2. \tag{5}$$

If the empirical distribution of $|r|$ is heavy-tailed with a Pareto-like upper tail $P(|r| > t) \sim t^{-\alpha}$, the contribution of the top fraction ε to (**??**) scales as

$$\mathbb{E}\Big[r^2\,\mathbf{1}_{|r|>t_\varepsilon}\Big] \;\propto\; \int_{t_\varepsilon}^{\infty} t^{2-\alpha-1}\,dt \;=\; \frac{t_\varepsilon^{2-\alpha}}{\alpha-2} \qquad (\alpha > 2), \tag{6}$$

which diverges as $\alpha \to 2^+$. Empirically the gen-3 corpus has α close enough to this regime that $<1\%$ of outliers contribute $>90\%$ of the mini-batch loss (`REPLACEMENT_PLAN.md` §4.C). The model-selection signal-to-noise ratio collapses.

Replacement loss: log-cosh. Define

$$\mathcal{L}_{\text{lc}}(r) \;=\; \log\cosh(r), \qquad \frac{d\mathcal{L}_{\text{lc}}}{dr} = \tanh(r). \tag{7}$$

The behaviour near zero is quadratic, $\mathcal{L}_{\text{lc}}(r) = \frac{1}{2}r^2 - \frac{1}{12}r^4 + \mathcal{O}(r^6)$, preserving the gradient on typical tracks. In the tails $\mathcal{L}_{\text{lc}}(r) = |r| - \log 2 + \mathcal{O}(e^{-2|r|})$ and $|d\mathcal{L}_{\text{lc}}/dr| \le 1$, capping the per-sample gradient contribution regardless of α. Substituting (**??**) into (**??**) replaces $r^{2-\alpha-1}$ by $r^{1-\alpha-1}$ in the heavy-tail integrand, restoring integrability for any $\alpha > 1$.

Selection metric. Model selection switches from $\mathcal{L}_{\text{MSE}}$ to the *median* $|\Delta x|$ on the held-out test split, with the 68%, 95% and 99% quantiles reported alongside. The median is a robust location statistic that is insensitive to the heavy tail of (**??**) and does not change under monotone transformations of the residual scale. `REPLACEMENT_PLAN.md` §4.C marks Fix L1 as *non-negotiable*: it must land before any further training run, because otherwise the heavy-tail population alone determines which checkpoint wins.

4 MLP modernisation (Fix M1)

The input-dimensionality regression. The gen-3 MLP was inherited verbatim from gen-2 with a 6-dimensional input $[x, y, t_x, t_y, q/p, \Delta z]$, missing the gen-3 z_0 coordinate. On the gen-3 corpus this produces an order-of-magnitude collapse: gen-2 `mlp_medium` reached 0.305 mm on forward-only data, whereas the inherited gen-3 `mlp_medium_v1` sits at 5.57 mm and `mlp_small_v1` at 18.41 mm (`REPLACEMENT_PLAN.md` §3, gen-3 row). The PINN_v2 and NeuralRK4 families had already absorbed z_0 via Fix H and signed Δz via Fix C2; the MLP recipe was the lone holdout.

Fix M1. The input layout becomes 7-dimensional,

$$\mathbf{x} \;=\; (x,\; y,\; t_x,\; t_y,\; q/p,\; z_0,\; \Delta z), \tag{8}$$

matching (**??**) and the existing `architectures.py` line-178 declaration `input_dim=7`. Two engineered features are appended:

$$\phi_1(\Delta z) \;=\; \log_{10}\!\left(\frac{|\Delta z|}{100\text{ mm}}\right), \qquad \phi_2(\Delta z) \;=\; \operatorname{sign}(\Delta z). \tag{9}$$

Why each feature is physically needed. *(i)* z_0*:* the LHCb dipole field profile is asymmetric about the VELO-exit plane (peak $|B_y| \approx 1.03\,\mathrm{T}$ at $z \approx 5007\,\mathrm{mm}$, `PROJECT_CONTEXT.md` §"Magnetic field map"). Two tracks with identical $(x, y, t_x, t_y, q/p, \Delta z)$ but different z_0 traverse qualitatively different field histories — short pre-magnet drift versus full magnet crossing — and produce different endpoint states. Without z_0 the MLP must average over these regimes and learns neither. *(ii)* $\operatorname{sign}(\Delta z)$*:* the gen-2 corpus had $\Delta z > 0$ always; gen-3 spans $\Delta z \in [-10000, +10000]\,\mathrm{mm}$ to support the Kalman-filter backward pass. The Lorentz ODE (**??**) is not symmetric under $z \to -z$ in the presence of the asymmetric $\mathbf{B}$, so forward and backward extrapolations are physically distinct maps. A symmetric 6-input layout cannot distinguish them. *(iii)* $\log_{10}|\Delta z|$*:* the dataset spans nearly three decades in $|\Delta z|$ ($25\,\mathrm{mm}$ to $10\,000\,\mathrm{mm}$). A log-scaled feature gives the network a basis adapted to this dynamic range, in the same spirit as the $\sigma_k/\Delta z$ non-dimensionalisation already used elsewhere (ISSUES §1).

5 PINN_v2 framework

Output parametrisation. Let $\mathbf{x}_0 = (x, y, t_x, t_y, q/p, z_0)$ be the initial state and $\zeta = (z - z_0)/\Delta z \in [0, 1]$ a normalised path parameter along the extrapolation step. The PINN_v2 trajectory is

$$u_\theta(\mathbf{x}_0, \zeta) \;=\; \mathbf{y}_{\text{base}}(\mathbf{x}_0, \zeta) \;+\; \zeta \cdot \delta_\theta(\,\tilde{\mathbf{x}}_0,\; \zeta\,), \tag{10}$$

where $\mathbf{y}_{\text{base}}$ is the straight-line baseline (initial state plus linear-in-ζ slope extrapolation), $\tilde{\mathbf{x}}_0$ is the non-dimensionalised initial state, and δ_θ is the network correction. The Fix C trunk of ISSUES §7C makes the encoder explicitly ζ-dependent, so (**??**) is *not* affine in ζ. The ζ-prefactor enforces the initial condition exactly: $u_\theta(\mathbf{x}_0, 0) = \mathbf{y}_{\text{base}}(\mathbf{x}_0, 0) = \mathbf{x}_0$.

Physics at training time only. The PDE residual

$$\mathcal{R}_{\text{Lorentz}}[u_\theta](\mathbf{x}_0, \zeta) \;=\; \partial_\zeta\, u_\theta(\mathbf{x}_0, \zeta) \;-\; \Delta z \cdot f_{\text{Lorentz}}\big(u_\theta,\; z_0 + \zeta\Delta z;\; q/p\big) \tag{11}$$

is evaluated on n_c collocation points $\zeta \sim \mathcal{U}(0, 1)$ drawn per sample per batch (Fix B, $n_c = 2$ in gen-3). The ∂_ζ derivative is computed by a single forward-mode JVP via `torch.func.jvp` under `torch.func.vmap` (Fix A), replacing the 32 reverse-mode backwards of the legacy implementation by one JVP per batch element. **Crucially, f_{Lorentz} and B appear only inside the training loss**; at inference time $\zeta = 1$ is fixed, the trajectory collapses to $u_\theta(\mathbf{x}_0, 1)$, and a single forward pass through δ_θ produces the output. The replacement criterion of Section **??** is preserved.

Loss decomposition. The total training loss is

$$\mathcal{L} \;=\; \mathcal{L}_{\text{data}} \;+\; \lambda_{\text{pde}}\, \mathcal{L}_{\text{pde}} \;+\; \lambda_{\text{ic}}\, \mathcal{L}_{\text{ic}}, \tag{12}$$

with $\mathcal{L}_{\text{data}}$ the log-cosh loss (**??**) on the endpoint prediction, $\mathcal{L}_{\text{pde}} = \langle \|\mathcal{R}_{\text{Lorentz}}\|^2_{\sigma/\Delta z} \rangle$ the componentwise non-dimensionalised mean of (**??**), and $\mathcal{L}_{\text{ic}} \equiv 0$ identically by construction (the ζ prefactor in (**??**) zeroes the correction at $\zeta = 0$); the λ_{ic} term is retained only for forward compatibility.

Fix P1 schedule. Linear warm-up $\lambda_{\text{pde}} : 0 \to \lambda^\star_{\text{pde}}$ over the first 10 epochs (extended from the gen-2 value of 5), followed by a held plateau, then a late-stage cosine decay $\lambda_{\text{pde}} : \lambda^\star_{\text{pde}} \to 0.1\,\lambda^\star_{\text{pde}}$ over the final 20% of training (`REPLACEMENT_PLAN.md` §4.B). The rationale is that the physics term is most useful as a trajectory-anchoring *regulariser* early in training; reducing its weight late allows the data term to polish the endpoint accuracy that matters at inference, without re-introducing the structural-floor pathology of the legacy PINN.

6 Allen integration constraints

The seven hard constraints (A1–A7) lifted from `REPLACEMENT_PLAN.md` §5 are reproduced in Table **??**, together with the verdict per surviving family. A4 is flagged explicitly as the open empirical question of Phase R2.

A4 is the open empirical question. The original A4 failure that motivated the move toward NeuralRK4 was diagnosed as an fp32 plus finite-difference artefact, not a structural property of point-prediction networks (ADR 0007 re-measurement, recorded in ADR 0009). A4 has *not* yet been re-measured on the true replacement candidates with the corrected fp64 + autograd methodology. Phase R2 (Section **??**) does this measurement before any further training is committed.

7 Roadmap (R1–R6 + R-X)

R-X contingency. If A4 cannot be met by point prediction alone, the diagnosis is that no point-prediction network can match the RK4 Jacobian. Two routes are documented in `REPLACEMENT_PLAN.md` §6 R-X, both of which preserve the replacement property:

1. *Jacobian co-supervision* — augment the training loss with

$$\mathcal{L}_J \;=\; \lambda_J \left\| \mathbf{J}_\theta(\mathbf{x}) - \mathbf{J}_{\text{RK45}}(\mathbf{x}) \right\|_F^2, \tag{13}$$

   with $\mathbf{J}_\theta = \partial f_\theta / \partial \mathbf{x}$ cheaply obtained via `torch.func.jacfwd`. The Jacobian is a property of the network alone; no field map enters at inference.

2. *Straight-line residual output parametrisation* — write

$$f_\theta(\mathbf{x}) \;=\; \big(\, x_0 + t_x \Delta z,\; y_0 + t_y \Delta z,\; t_x,\; t_y,\; q/p \,\big) \;+\; \Delta_\theta(\mathbf{x}), \tag{14}$$

   so the network learns only the deviation from a closed-form linear baseline. The straight-line Jacobian is constant and known, and the network's Jacobian contribution is small by construction, easing A4. No field map and no f_{Lorentz} enter (**??**); the criterion of Section **??** is preserved.

8 What changed from prior documents

The re-anchoring of 2026-05-19 supersedes a sequence of prior design decisions. Cross-references are to the `CLEANUP_LIST.md` where the corresponding code/doc action is itemised.

- **ADR 0002 / 0003 / 0007 superseded by ADR 0009.** The progression "hybrid NeuralRK4 with learned corrector → pure classical RK4 with $n_{\text{rk\_steps}} = 2$, corrector OFF" is retired. Neither endpoint satisfies the canonical replacement goal (ADR 0009).

- **`NeuralRK4` demoted.** Moved from "M1 deployment candidate" to "research baseline / F2–F3 audit reference". Checkpoints relocated to `trained_models/_archive_hybrid/`; configs to `configs/_archive_hybrid/`; module docstring updated (CLEANUP_LIST §§2.1, 3.1, 3.2).

- **Gen-3 MLP collapse diagnosed.** Root cause is the missing z_0 and the symmetric input layout under signed Δz (Section ??). `mlp_{small,medium}_v1` marked `_v1_broken`; retraining as `_v2` per Phase R3 (CLEANUP_LIST §2.3).
- **Gen-3 `README.md` re-anchored.** The M1 results table gains a "Replacement?" column; the thesis-section sketch reverses direction — the open question is now whether scaled PINN_v2 closes the 0.1 mm gate, not whether NeuralRK4 supplants PINN_v2 (CLEANUP_LIST §§4.1, 4.2).
- **`For_Allen/PLAN.md` re-anchored.** Eight phases re-pointed from `nrk4_tiny_1step_v1` to "the §R4 PINN_v2 winner". The Allen surface `NRKExtrapolator.cuh` is preserved as the right abstraction layer; its inner loop will be replaced by a generic `forward_pass(weights, state)` call in a follow-up MR (CLEANUP_LIST §§1.1, 1.3, 4.3).
- **Loss-metric reform mandated project-wide.** MSE is no longer a valid selection metric for any candidate; log-cosh + median $|\Delta x|$ is mandatory (Section ??; CLEANUP_LIST §0 prerequisites).

References

PROJECT_CONTEXT `PROJECT_CONTEXT.md` — canonical project goal and architecture taxonomy.

REPLACEMENT_PLAN `experiments/gen_3/REPLACEMENT_PLAN.md` — §§1, 2, 4, 5, 6 (R1–R6, R-X).

CLEANUP_LIST `experiments/gen_3/CLEANUP_LIST.md` — off-goal artefact inventory.

ADR 0009 `For_Allen/docs/decisions/0009-replacement-goal-restated.md` — controlling decision record.

ISSUES (gen-2) `experiments/gen_2/ISSUES_WITH_PINNS_AND_RK_PINNS.md` — legacy PINN / RK_PINN pathology audit.

architectures.py `experiments/gen_3/models/architectures.py` — `MLP` (line 178), `PINN_v2` (line 232), `NeuralRK4` (line 415).

gen-3 M1 results `docs/reports/gen3_m1_results.tex` — LaTeX preamble convention; M1 headline numbers.

gen-3 protocol `docs/reports/gen3_protocol.tex` — notation conventions (q/p, Δz, $\mathbf{y}$, $\mathbf{J}$, $\mathcal{L}$).

Companion (results) `docs/reports/gen3_replacement_results_2026-05-19.tex` — empirical companion to the present document.

Table 2: Allen Stage-1 integration constraints A1–A7 and their hard limits. Verdicts per replacement family. "?" = untested, awaiting Phase R2.

ID	Constraint	Hard limit	MLP	P
A1	Constant memory at inference; no per-event allocation	strict	✓	
A2	Weight export to fp32 & fp16 with deterministic round-trip	bit-exact reload	✓	
A3	Total trainable weights $\leq$ 64 kB (GPU shared-mem budget)	hard	✓ (width $\leq [256, 256]$)	✓ up to $[256, 256]$; the wider $[512, 256, 256]$ va
A4	$\Vert \mathbf{J}_{\text{model}} - \mathbf{J}_{\text{RK45}} \Vert_F / \Vert \mathbf{J}_{\text{RK45}} \Vert_F < 0.05$, max off-diagonal rel-err < 0.20	hard (Kalman)	**?**	
A5	fp32 inference reproduces Python output t	hard	✓ (proven via `TrackMLPExtrapolator.cpp`)	✓ (same fo

Table 3: The six-phase replacement roadmap from `REPLACEMENT_PLAN.md` §6, plus the R-X contingency.

Phase	Exit criterion	Artefact produced
R1 (Fix L1)	Published median, 68%, 95%, 99% $\lvert\Delta x\rvert$ and $\lvert\Delta\text{slope}\rvert$ for all M1 checkpoints under log-cosh selection	Re-evaluated metrics tables; updated `train.py` / `eval.py`
R2 (A4 gate)	A4 measured at fp64 on `mlp_medium_v1` and `pinn_v2_small_v1` against the cached RK45 Jacobian reference	`evaluate_a4(model, test_states)` utility; pass/fail decision
R3 (Fix M1)	At least one MLP $<$ 0.5 mm median $\lvert\Delta x\rvert$ on the 10 M corpus with the 7-dim + engineered-feature input	`mlp_{small,medium,large}_v2` checkpoints
R4 (Fix P1)	At least one PINN_v2 $<$ 0.10 mm median $\lvert\Delta x\rvert$, passing A4	PINN_v2 width sweep; **the Allen candidate**
R5 (export + bridge)	Bit-bound smoke test green on the R4 winner; `NRKExtrapolator.cuh` re-purposed to call `forward_pass(weights, state)`	Re-anchored `For_Allen/PLAN.md`; weight blob
R6 (Allen MR + Moore)	Throughput improvement in `hlt1_pp_default`; accuracy gates hold in Moore physics tests	Production MR
R-X (contingency)	Triggered if both MLP and PINN_v2	See below